CS 237: PROBABILITY IN COMPUTING

INSTRUCTORS: TIAGO JANUARIO, NATHAN KLEIN

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LECTURE 1

- Course information
- Introduction to probability theory
- Sample Spaces and Events
- Examples

INSTRUCTORS

Nathan Klein

nklei1@bu.edu

https://nathan-klein.github.io/

Tiago Januario

januario@bu.edu

https://cs-people.bu.edu/januario/

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STAFF MEMBERS

- Teaching Fellow:
  - Ta Duy Nguyen
- Teaching Assistants:
  - Daniel Matuzka
  - Steve Choi
- Course Assistants:
  - Vi Tjiong
  - Sarah Yuhan
  - Yoon Oh

Check the class website for Office Hours and more

https://srtjanuario.github.io/2026.spring.cs237/

COMMUNICATION

- https://piazza.com/bu/spring2025/cascs237
- Use the available "folders" for each question
- Before asking your question, think about how you can frame it to benefit you and other students in the course.
- Bad questions exist and will hurt our ability to answer you efficiently.
- Check our Google Calendar for Office hours

PREREQUISITES

- CS 131: Combinatoric Structures
- MA 123: Calculus I
- CS 111: Introduction to Computer Science 1
- Good working knowledge of:
  - Sets, counting, proofs (induction, contradiction, etc.)
  - Differentiation and integration
  - Python programming

COURSE STRUCTURE

- Lectures: Tuesdays and Thursdays at KCB 106
- Discussions: Fridays at MCS B33
- Attendance in lectures and discussion is mandatory
- The two lecture sections of the course will be treated as one class.
  - **You must attend the section you are signed up for.**
- The content of the two lectures is identical, and assignments will be shared.
- If you missed a lecture, check the annotated slides on Piazza.

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SCHEDULE AND TEXTBOOK

Date / Lec	Agenda (Topics, Readings, Homework)	Instr.
Lec 1 Tuesday Jan 20	Course information Tips to succeed Random experiments Sample spaces, events Read: P 1.1, P 1.2, OB 1B Do: Jupyter Lab, Your first Jupyter Notebook, Collaboration & Honesty Policy	TJ
Lec 2 Thurday Jan 22	Probability function Symmetry Probability axioms Read: LLM 17.1, P 1.3.1-1.3.3 Do: hw01 out	TJ
Lec 3 Tuesday Jan 27	Probability rules Computing probabilities Read: LLM 17.3, LLM 17.5, P 2	TJ
Lec 4 Thurday Jan 29	Tree diagrams The Monty Hall problem Read: LLM 17.2, LLM 18.1.2 Do: hw02 out	TJ
Lec 5: Quiz 1 Tuesday Feb 3	Continuous Probability Spaces Anomalies with Continuous Probability Read: P 1.3.5 Watch: Video, Why "probability of 0" does not mean "impossible"	TJ

Textbook: Mathematics for CS (probability part)

https://cs-people.bu.edu/aene/cs237fa21/mcs.pdf

ATTENDANCE AND PARTICIPATION

Required in-class software: **Top Hat Pro platform**

- You will get full participation points if you answer at least 85% of the possible Top Hat questions.
- You will get full attendance points if you attend at least 85% of the discussion labs.
- If you end up with x% points, where x < 85, you will get x/85 of the points.
- We will start using Top Hat in the second week of classes.

https://tophat.com

HOMEWORK POLICY

- Deadline: Thursdays, at 09:00 PM, with a 3-hour grace period.
- No extensions, except for some students with formal accommodation.
- You may use up to 3 grace periods without penalty.
  - After that, each late submission incurs a 1% grade penalty.
- The lowest homework grade will be dropped after penalties are applied.

gradescope

SUBMIT YOUR HOMEWORK THE RIGHT WAY

- You are responsible for submitting **one single PDF file** with high-quality images of your solutions.
- Illegible submissions will receive a 0 grade
- We highly recommend Dropbox to scan your homework before uploading it.
- Select the correct pages on Gradescope for each problem (solved or not) to avoid a 10% homework penalty.
- In cases where you do not have a solution to submit for a specific problem, write a brief note such as "No solution provided".

LATEX

The Not So Short Introduction to $\LaTeX\ 2_\varepsilon$

Or $\LaTeX\ 2_\varepsilon$ in 139 minutes

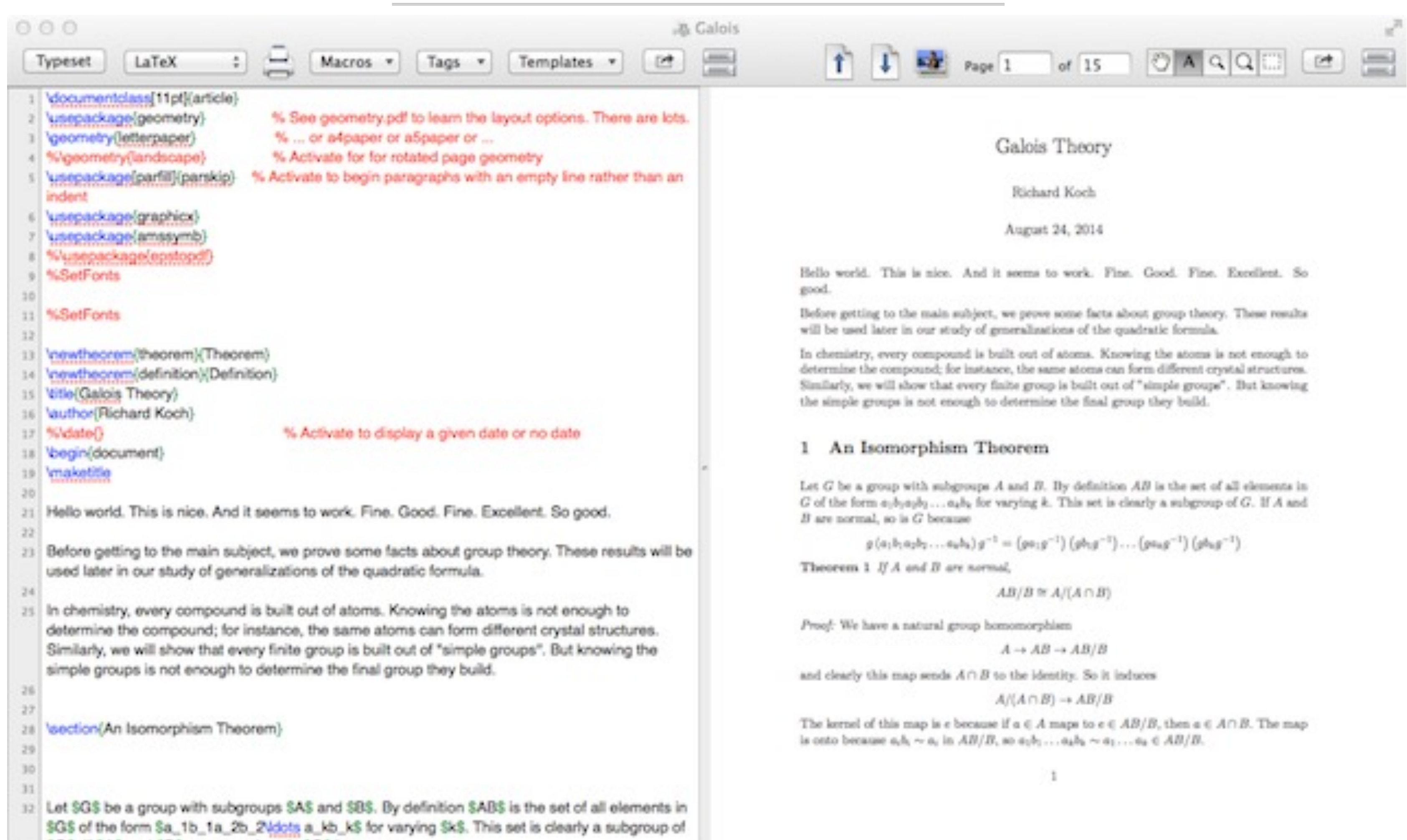
by Tobias Oetiker

Hubert Partl, Irene Hyna and Elisabeth Schlegl

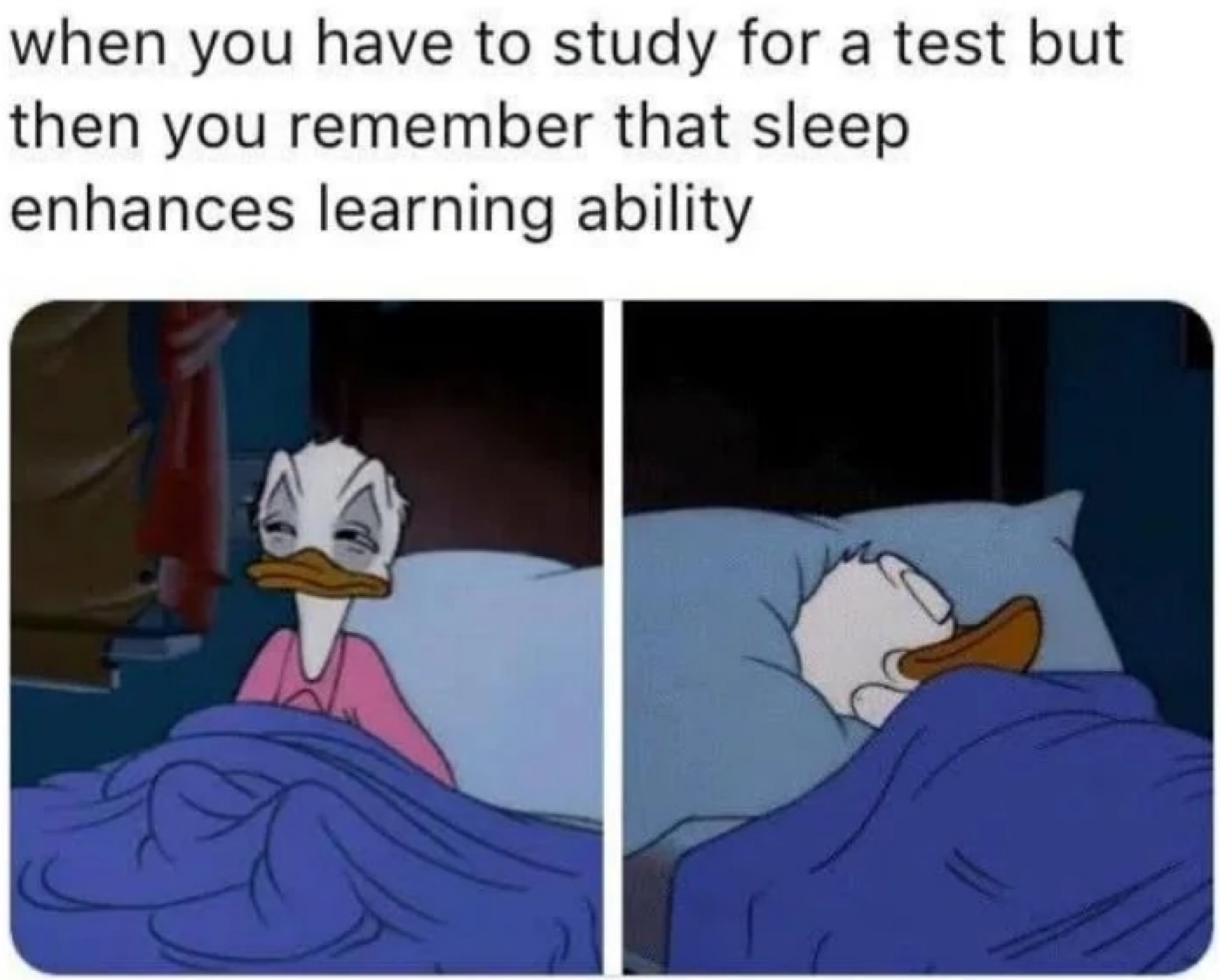
Version 6.3, March 26, 2018

LATEX

- Many editors: TeXShop, TeXStudio, [overleaf.com, …]



EXAMS



[Sleep tips for exam success]

- Midterm and Final exams will consist of problem-solving and short questions about the material.
- Each exam duration and its location are given in the course schedule.
- Makeup exams will only be given in documented cases of serious illness.
- The content of the Final exam is cumulative.

QUIZZES

- There will be four quizzes based on flashcards that will be posted weekly on Piazza.
- Quizzes are all cumulative and will be simple, short answers based on the flashcards.
- Quizzes will be held at the beginning of class and will last 10-15 minutes.
- Your lowest quiz score will be dropped.

https://apps.ankiweb.net

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GRADING

- Attendance (5%) and Participation (5%)
- Quizes (10%)
- Homework: problem sets posted every week (20%)
  - Math proofs and programming problems
- In-class midterm exam (25%)
- Final exam (35%)
- Check the course webpage for times and locations

PYTHON

- We will use Python 3 and Jupyter Lab
- https://jupyter.org/try-jupyter/lab/

REGRADE POLICY

- Regrade requests can be submitted up to one week (7 days) after grades have been posted (except the final exam).
- You must request a regrade via Gradescope, NOT through email.
- When we regrade a problem, your score may go up or down.

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TIPS FOR THE COURSE

- Concepts in this course take some time to sink in: be careful not to fall behind.
- Prepare for each lecture by reviewing material from the previous lecture and doing assigned reading.
- Be active in lectures, discussions, and on Piazza.
- Take advantage of office hours.
- Solved exercises in Pishro-Nik's textbook

TIPS FOR THE COURSE

- Study with a friend: do exercises and quiz each other.
- Allocate lots of time for the course: comparable to a project course, but spread more evenly.
- Start working on HW early and solve it over multiple days.
- You can work in groups (up to 3 people), but spend at least 30 minutes thinking about it on your own before your group meeting.

COLLABORATION AND HONESTY POLICY

- https://cs-people.bu.edu/januario/teaching/cs237/collaboration-policy.pdf
- Read, sign, and submit it to Gradescope.
- Discuss with classmates (strongly encouraged!)
- Write up in your own words, acknowledge people you worked with
- Do not share written work, and write your own code!
- Do not submit anything you cannot explain to the course staff

WHAT IS CS 237 ABOUT?

Fundamentals of Probability

- What is a random process and how can we model it and analyze it?
- Basic probability and events
- Conditional probability and independence
- Discrete and continuous random variables and distributions
- Expectation and variance

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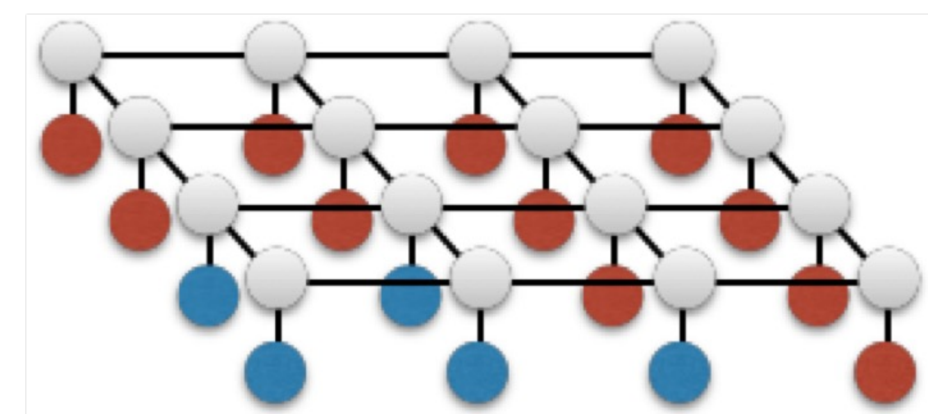
WHAT IS CS 237 ABOUT?

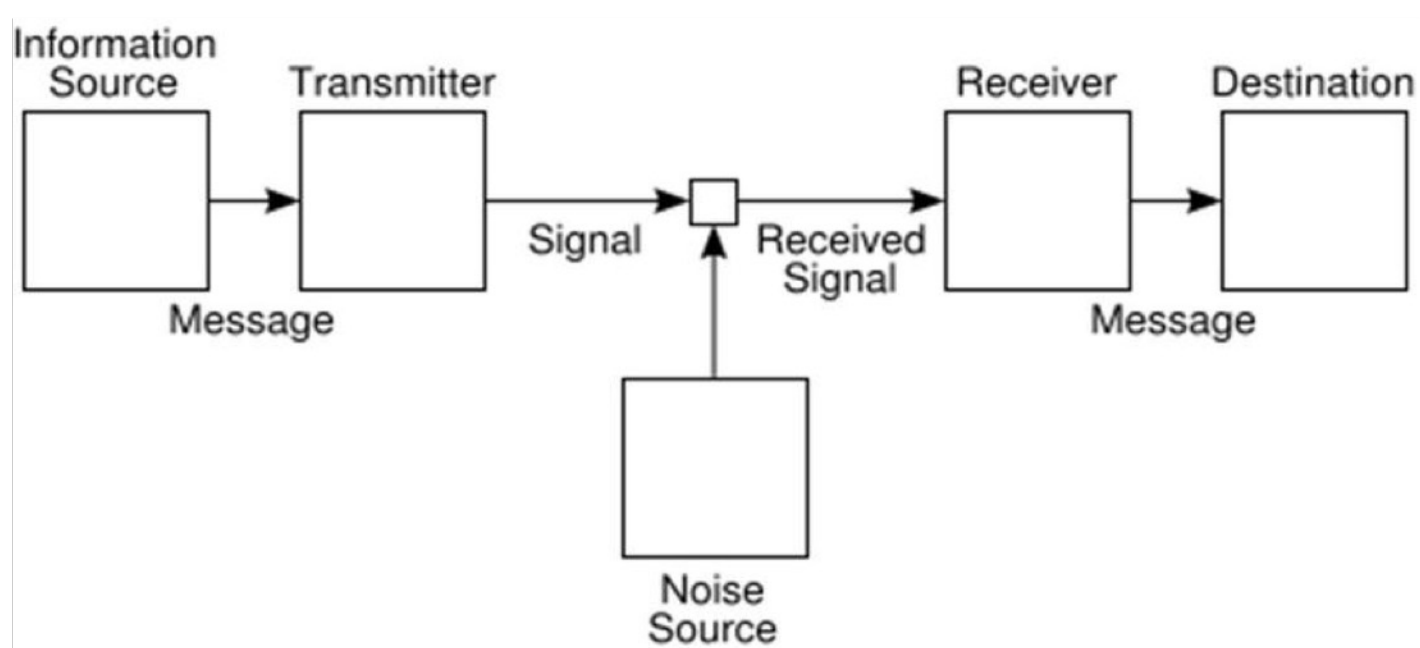
Fundamental Tools for CS

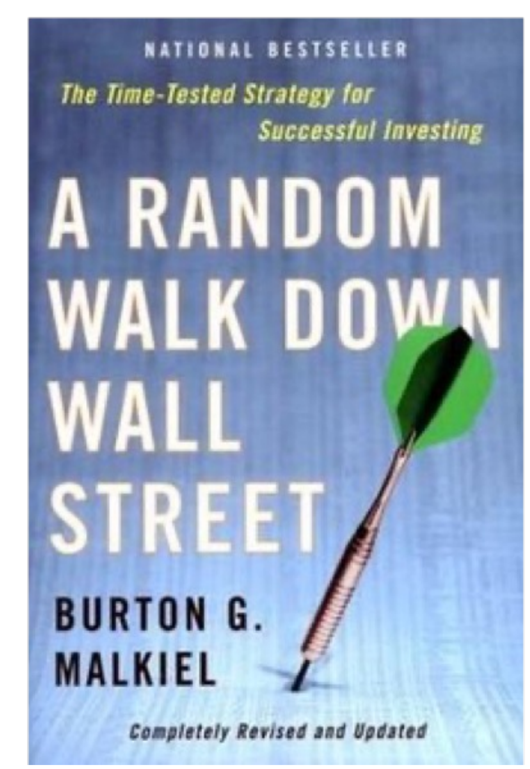
- Estimation by sampling
- Probabilistic analysis via concentration inequalities
- Probabilistic Data Structures
- Randomized algorithms

PROBABILITY IS UBIQUITOUS

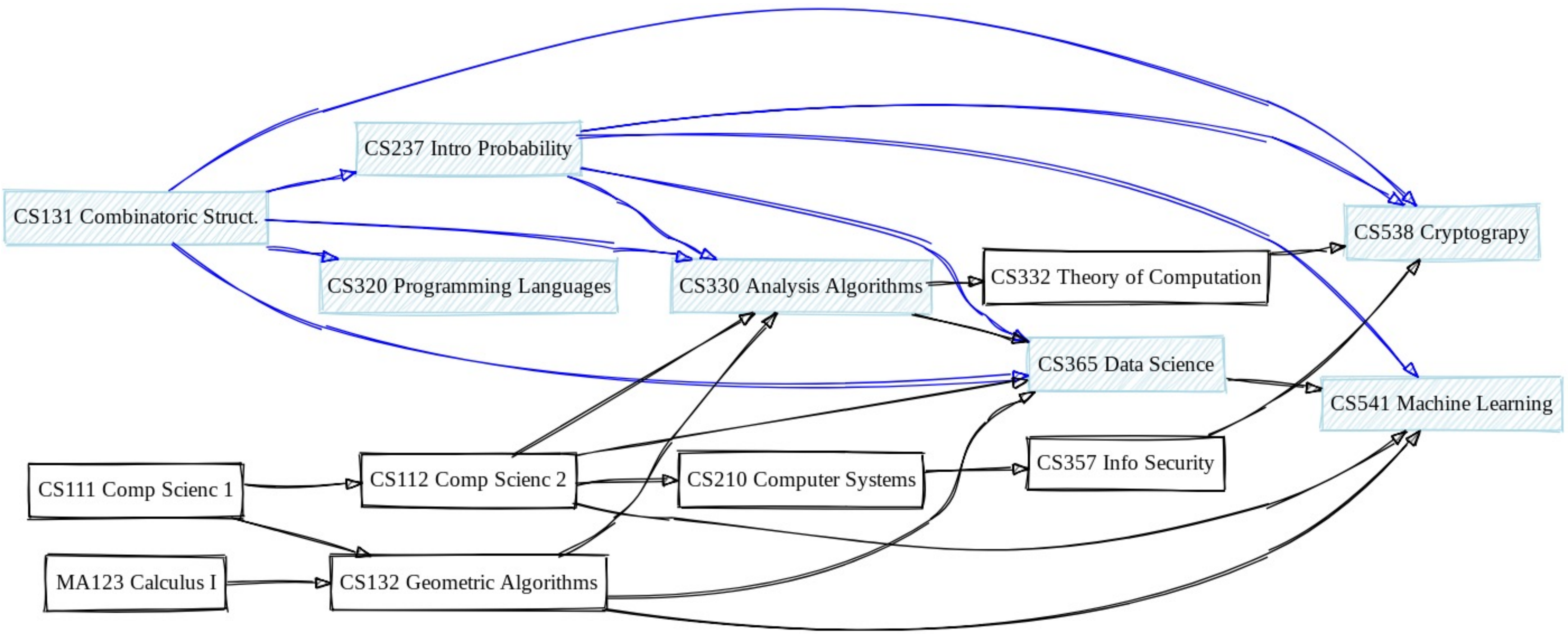
Computing Statistics Engineering Economics/Finance Linguistics







PROBABILITY IN THE CS CURRICULUM



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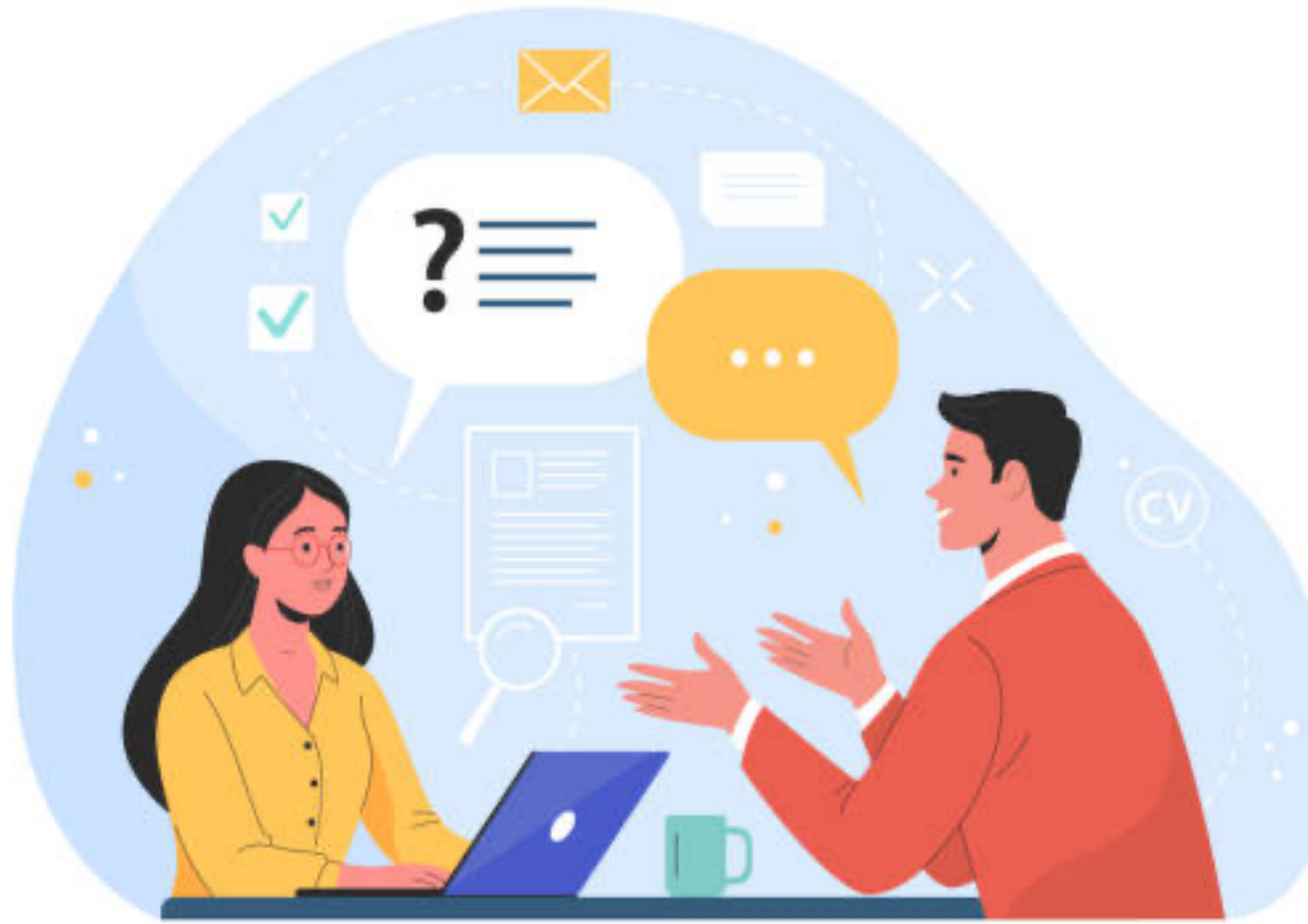
WHAT ABOUT CS 237 MATERIAL ON JOB INTERVIEWS?

- InterviewBit
- Indeed
- Interview Query
- Nick Sigh
- ML Stack Café
- Strata Scratch
- Towards Data Science



QuantProof

LET'S ROLL SOME DICE

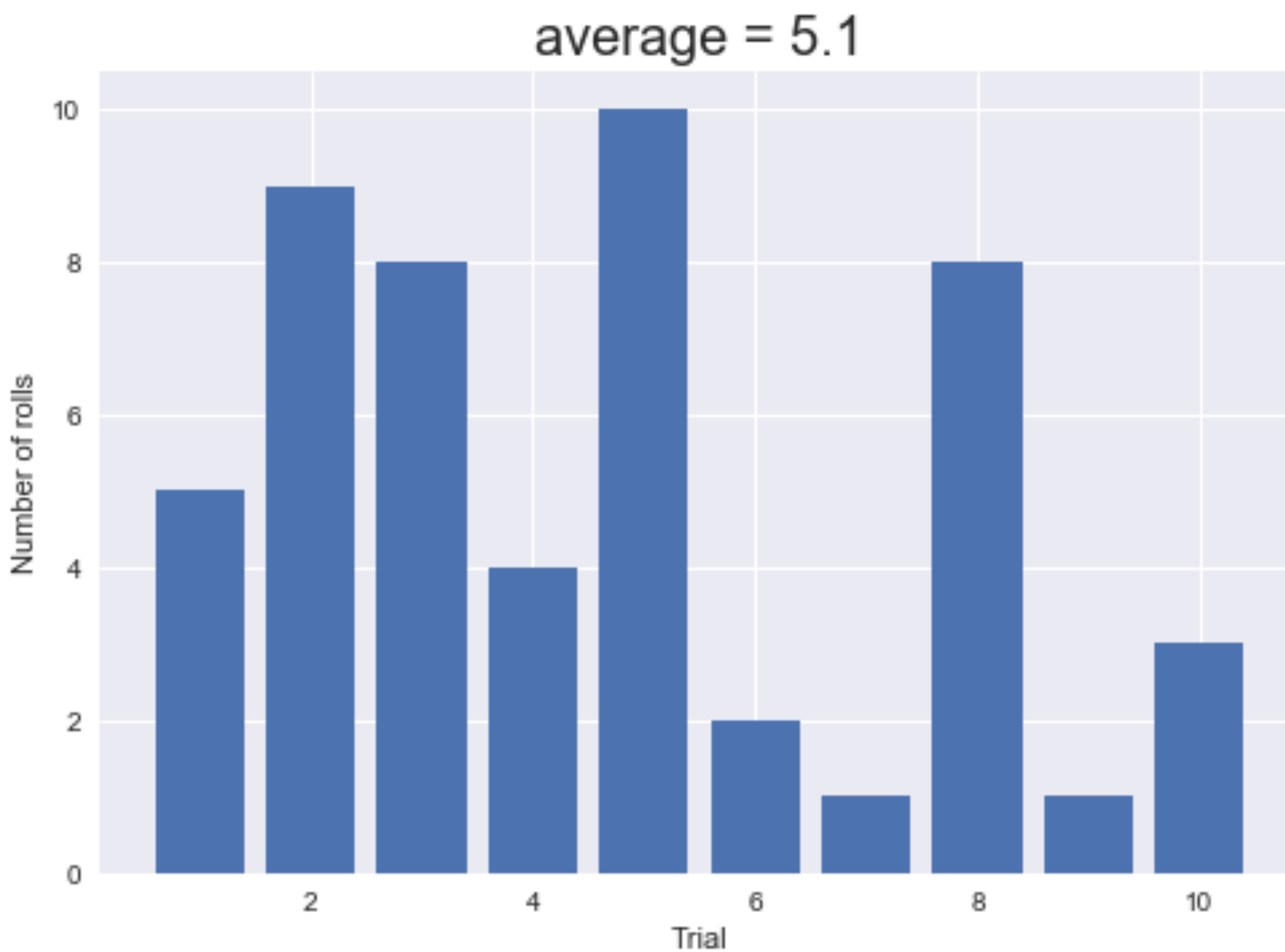
- Take a fair 6-sided die 
- Roll the die until we get a 6
- What is the expected number of rolls?

- Let's run an experiment using Jupyter Lab

LET'S ROLL SOME DICE

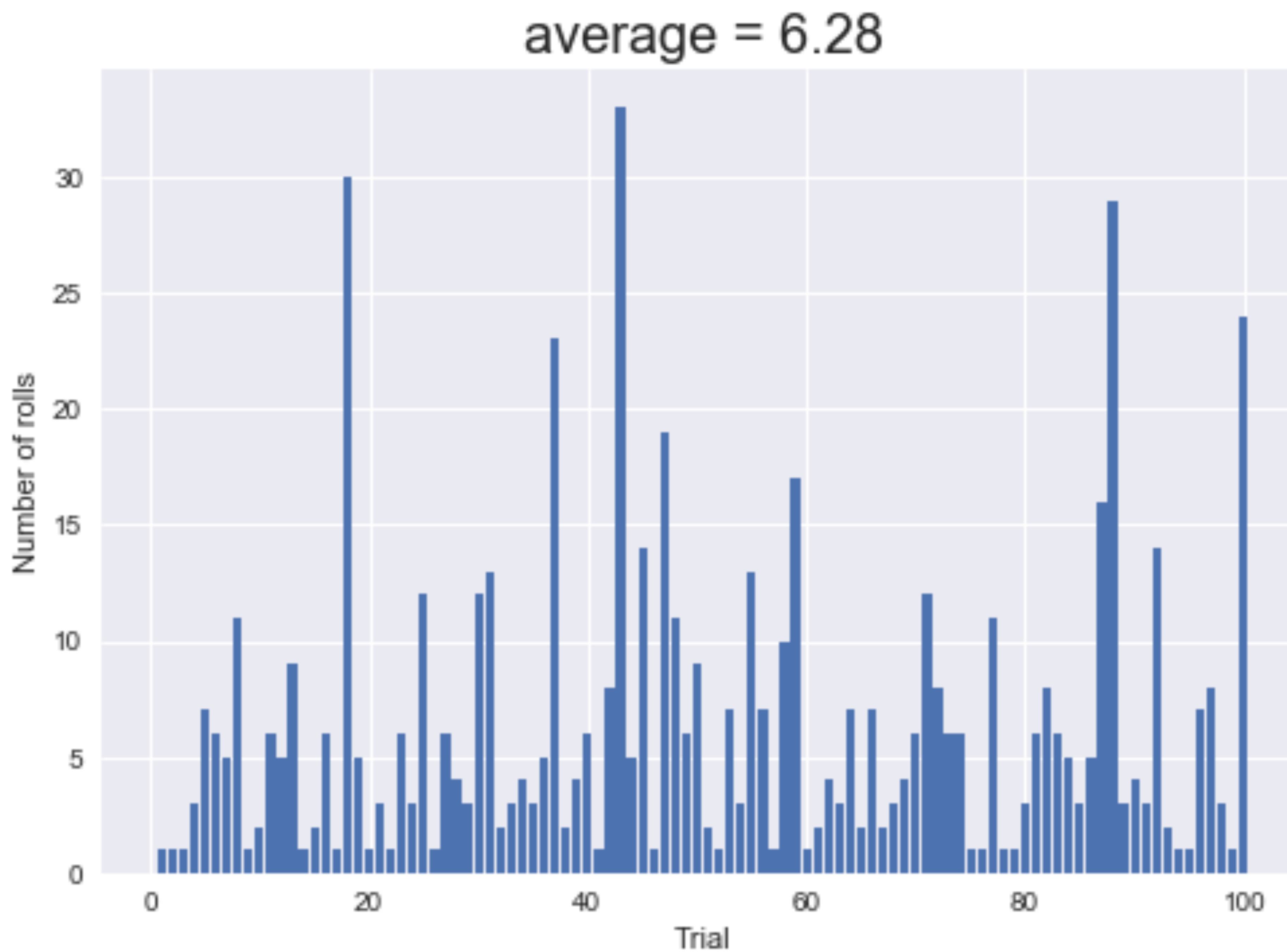
```
# a single experiment
def single_trial():
  num_rolls = 0
  while True:
    num_rolls = num_rolls +1
    die_roll = random.randint(1,6) #fair die roll
    if die_roll == 6:
      break
  return num_rolls

# perform N trials
N = 10
rolls = []
trial = [i+1 for i in range(N)]
for i in range(N):
  num_rolls = single_trial()
  rolls.append(num_rolls)
```

LET'S ROLL SOME DICE



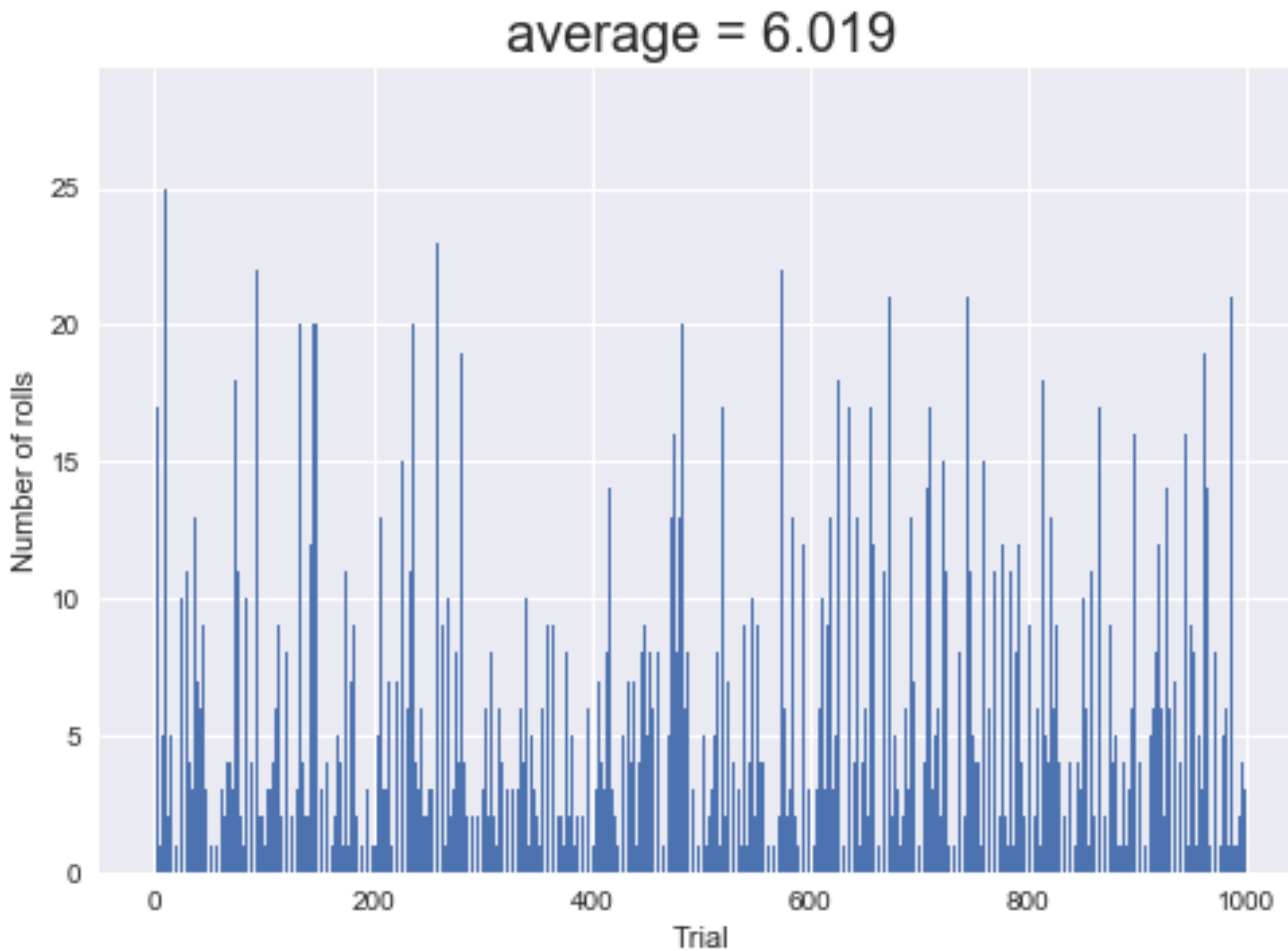
10 trials

LET'S ROLL SOME DICE



100 trials

LET'S ROLL SOME DICE



1000 trials

LET'S ROLL SOME DICE

- Fair 6-sided die
- Roll the die until we get a 6
- What is the expected number of rolls?
- Correct answer = 6

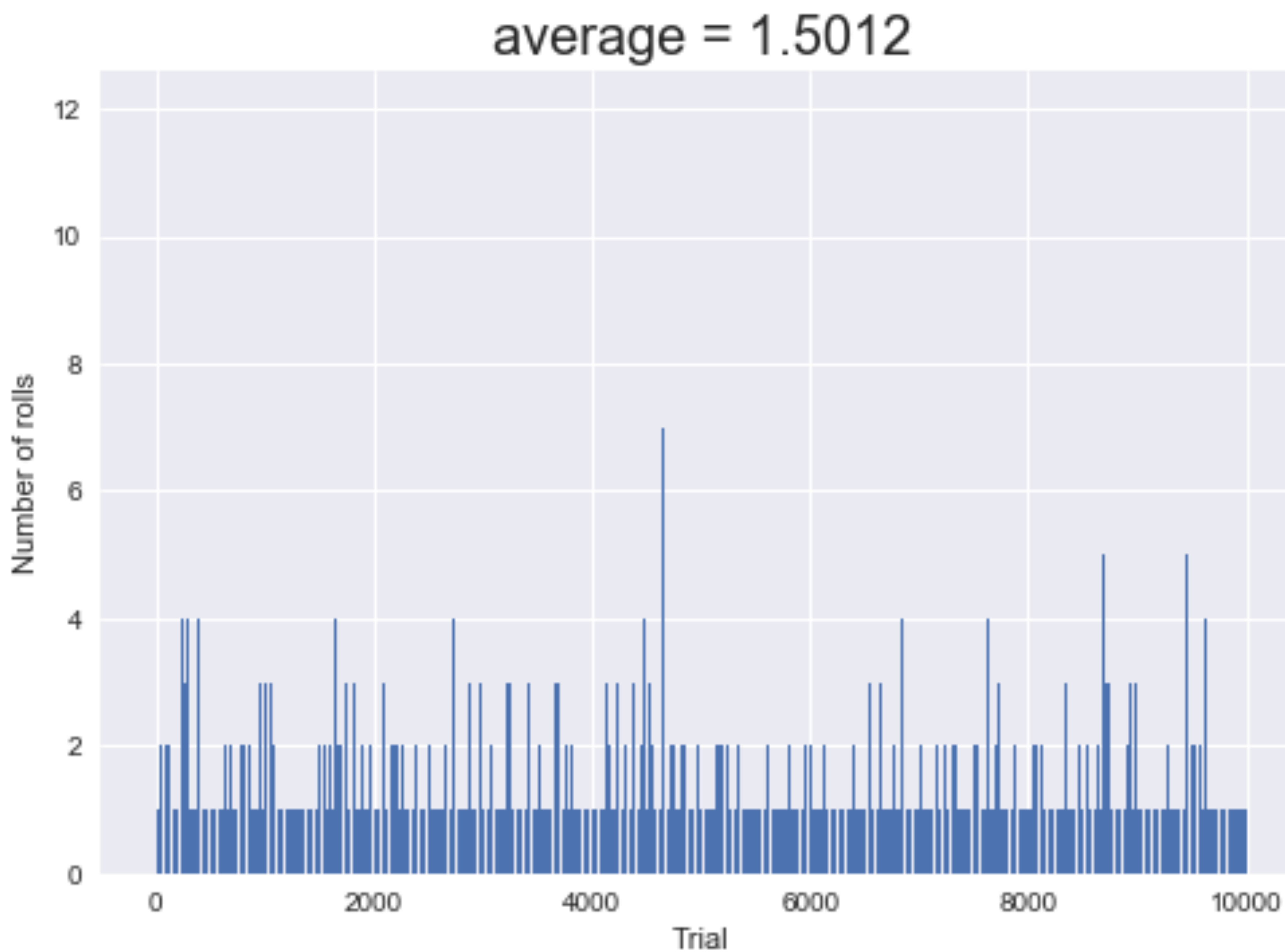
LET'S ROLL SOME DICE

- Fair 6-sided die
- Roll the die until we get a 6
- What is the expected number of rolls given that all rolls gave even numbers?

LET'S ROLL SOME DICE

```
# a single experiment
def single_trial():
  num_rolls = 0
  while True:
    num_rolls = num_rolls +1
    die_roll = random.randint(1,6) #fair die roll
    if die_roll % 2: # restart the experiment
      num_rolls = 0
    if die_roll == 6:
      break
  return num_rolls

# perform N trials
N = 1000
rolls = []
trial = [i for i in range(N)]
for i in range(N):
  num_rolls = single_trial()
  rolls.append(num_rolls)
```

LET'S ROLL SOME DICE



LET'S ROLL SOME DICE

- Fair 6-sided die
- Roll the die until we get a 6
- What is the expected number of rolls given that all rolls gave even numbers?
- Correct answer = 1.5

PROBABILITY THEORY

- Introduce three self-evident and indisputable properties of probability (the axioms)
- Develop the mathematical theory of probability from these axioms

Andrey Kolmogorov
[1903 - 1987]

"The theory of probability as a mathematical discipline can and should be developed from axioms in exactly the same way as geometry and algebra."

TYPICAL STATEMENTS ABOUT PROBABILITY

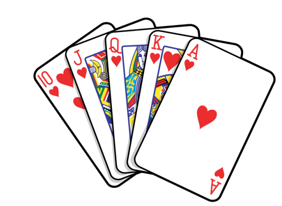
1. The probability that a randomized algorithm for checking polynomial identities accepts when the input is not an identity is at most 1/100.
2. The chance of getting a flush (that is, all cards of the same suit) in a 5-card poker hand is about 2 in 1000.
3. The chance of precipitation today in Boston is 20%.

Each such statement is implicitly talking about a random experiment

- either constructed by us, as in (1) and (2)
- or used by us to model the world, as in (3)

PROBABILITY: RANDOM EXPERIMENT

- Random experiment: a repeatable procedure
  - Pick an integer from the set $\{1, \ldots, 100\}$
  - Toss a coin
  - Toss a coin 3 times
  - Roll two dice
  - Pick a 5-card hand out of a deck of cards
  - Observe the number of goals in a soccer match between robots



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PROBABILITY: SAMPLE SPACE

- Outcome: result of the experiment
- Sample space Ω: set of all possible outcomes
  - Pick an integer from the set $\{1, \ldots, 100\}$

    $\Omega = \{ i \mid 1 \leq i \leq 100, \text{ and } i, j \in \mathbb{N} \}$
  - Toss a coin

    $\Omega = \{H, T\}$, $\Omega = \{\text{Heads}, \text{Tails}\}$
  - Toss a coin 3 times

    $\Omega = \{HHH, HHT, HTH, THH, TTH, THT, HTT, TTT\}$

    $|\Omega| = 2 \cdot 2 \cdot 2 = 8$

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PROBABILITY: SAMPLE SPACE

- Outcome: result of the experiment
- Sample space Ω: set of all possible outcomes
  - Roll two dice

    $\Omega = \{(i,j) \mid 1 \leq i,j \leq 6 \text{ and } i,j \in \mathbb{N}\}$

  - Pick a 5-card hand out of a deck of cards (the order of cards doesn't matter, so {5♥, 5♦, 5♠, Q♥, Q♣} and {5♦, 5♥, Q♥, 5♠, Q♣} are the same hand)

    Ω is the set of all subsets of five cards, $|\Omega| = \dfrac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5!} = \binom{52}{5} = \dfrac{52!}{5!\,(52-5)!}$

  - Observe the number of goals in a soccer match between robots

    $\Omega = \{0, 1, 2, \ldots\}$ or $\Omega = \mathbb{N}$

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PROBABILITY: EVENT

- **Event:** a subset of the sample space

  maximal

  (that is, a set of outcomes)

  - Experiment: toss a coin 3 times

    Event A: get at least 2 heads

    $A = \{$ HHT, HTH, THH, HHH $\}$

  - Experiment: roll two 6-sided dice $\Omega = \{(a, b) \mid 1 \leq a, b \leq 6\}$

    1st die

    2nd die

    Event B: the sum of the two numbers rolled is 11

    $B = \{ (5,6), (6,5) \}$

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ASK THE AUDIENCE

Experiment: toss a coin 3 times. Which of the following is the event "exactly 2 heads"?

has 3 heads

- $E_1 = \{HHT, HTH, THH, HHH\}$
- $E_2 = \{HHT, HTH, THH\}$
- $E_3 = \{HTH, THH\}$ ← its not maximal

A. E_1

B. E_2

C. E_3

D. Both E_2 and E_3 are correct

ASK THE AUDIENCE

Experiment: toss a coin 3 times

- Event $E = \{HTH, HHT, THH\}$

Which of the following describes the event E?

A. "exactly one head"

B. "exactly one tail"

C. "at most one tail" → 0 or 1 tails

D. None of the above

EVENTS: Homework: What happens if $A \cap B = \emptyset$?

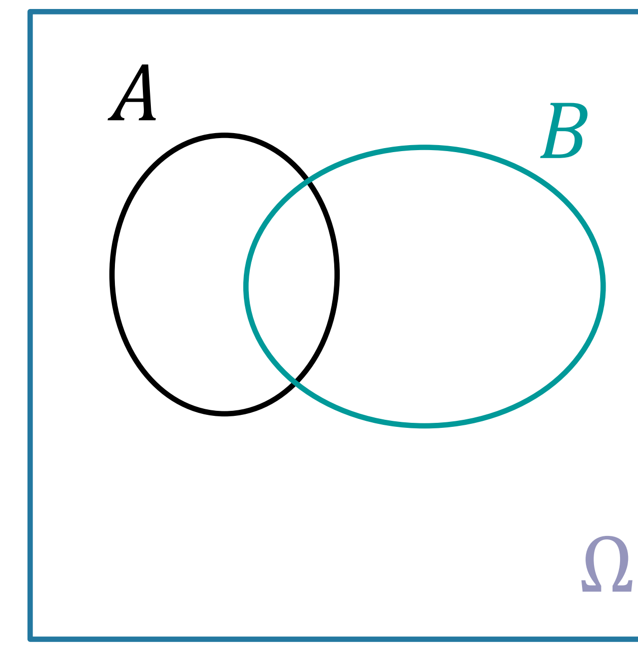
- Events are *sets* of outcomes
- We can combine events using set operations

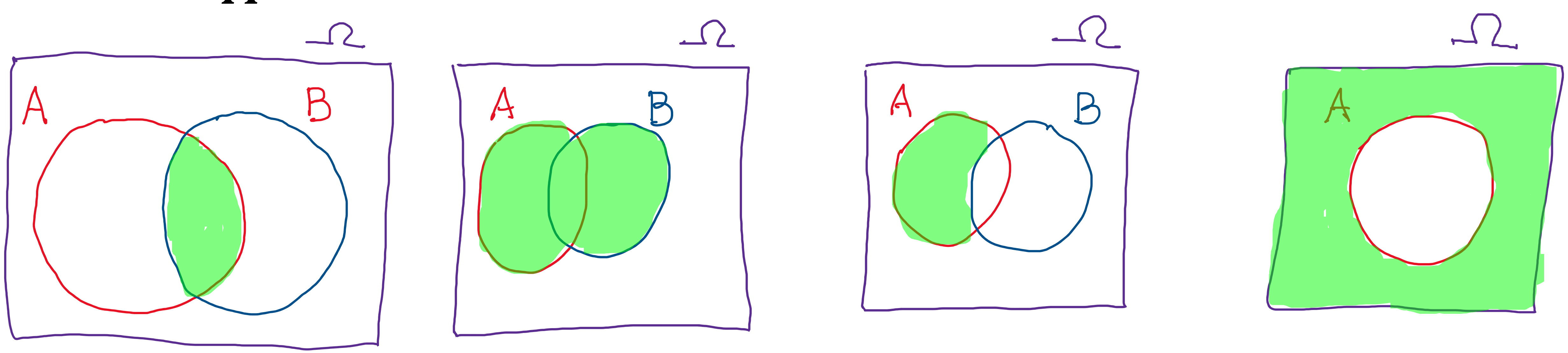
$A \cap B$: the event that both A and B occurred

$A \cup B$: the event that A or B occurred

$A \setminus B$: the event that A occurred but B did not

$\overline{A}$: the event that A did not occur





CHECKLIST

- Go to the course website
- Browse course materials
- Make sure you understand the course policies
- Sign up on Piazza
- Sign up on Gradescope
- Install LaTeX and Python